Dynamic Tutor Agent for K-12 Students with ADHD

With Reinforcement Learning

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Defining the Problem: Markov Decision Process Model

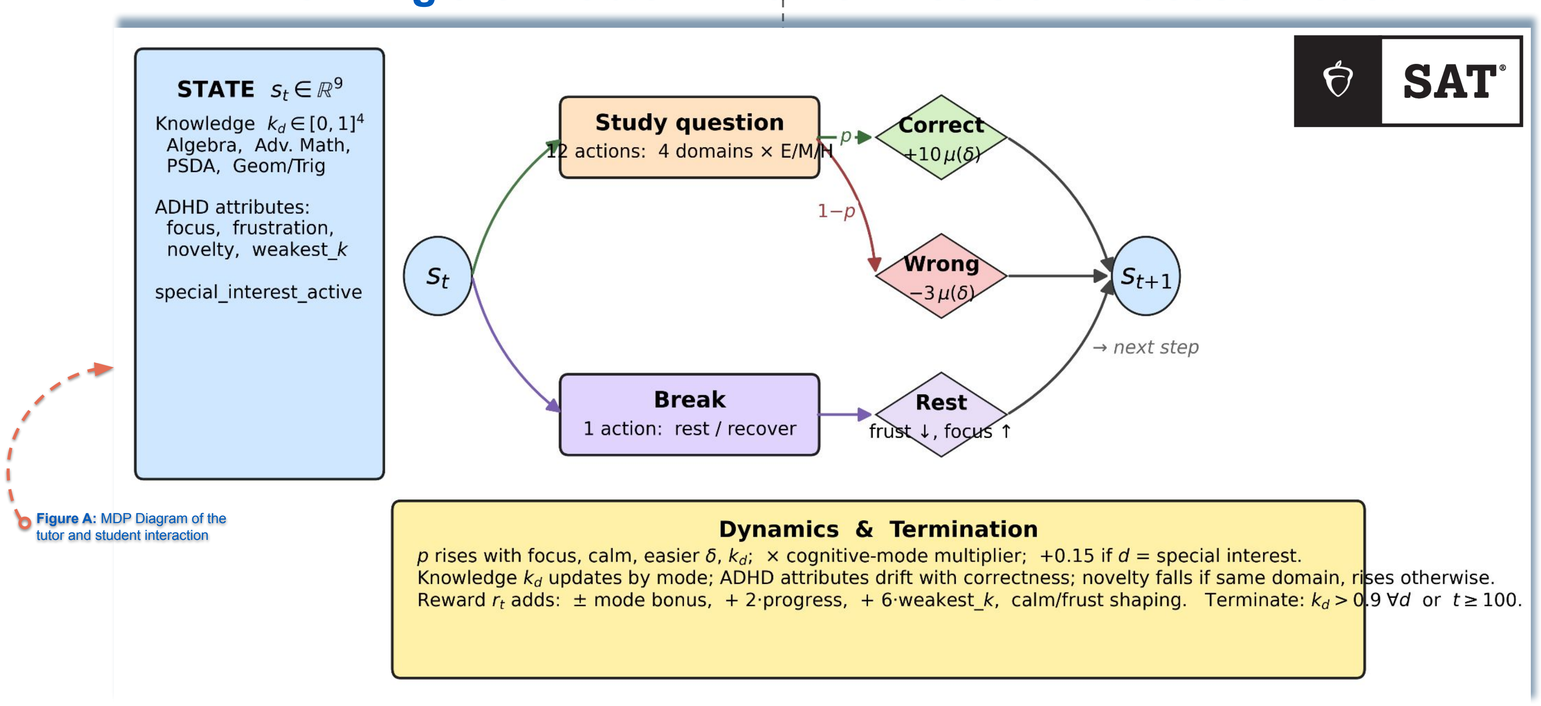


Figure A: MDP Diagram of the tutor and student interaction

Abstract

Attention-Deficit/Hyperactivity Disorder (ADHD) affects approximately **5-7%** of school-aged children **worldwide**. It is characterized by inattention, hyperactivity, and impulsivity, which frequently **hinder academic and social success** (National High School Journal of Science, 2025).

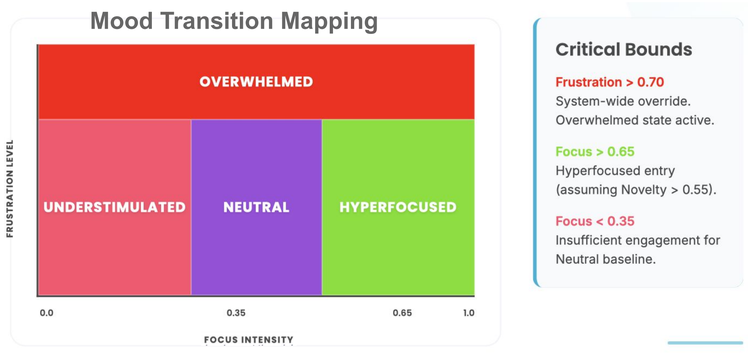
Students with ADHD face unique challenges in a traditional one-size-fits all learning program. This project develops a dynamic tutor that guides a student to prepare for the SAT math exam using real SAT questions.

Modeling ADHD Behavioural Traits

We modeled our student environment to have ADHD attributes that produce stochastic learning behaviours.

- frustration levels
- focus intensity
- special interest
- feeling of novelty
- rolling progress

these attributes determine the student's cognitive mode, which governs their ability to learn and perform.



Training

The tutor agent was trained with

- **25000 iterations** over
- **1000+ real SAT** math questions across
- **4 Domains** and
- **3 Difficulty Levels**.

Methods

1. We **model** the problem as a MDP where the agent observes the student's cognitive state and selects the next question domain and difficulty at each timestep.
2. To **train** the dynamic tutor, we implemented 2 **deep RL** algorithms from scratch.
3. We **measured and compared** the dynamic tutor results against a control tutor with a fixed policy.

Control Tutor: Fixed Policy

- Selects questions using **hard-coded** thresholds on student mental state and recent performance

Dynamic Tutor: DDQN, PPO algorithms

- **DDQN** is ideal for approximating Q-values for discrete action spaces like choosing from a set of domains and questions for the student.
- **PPO** contrasts DDQN by learning the policy directly over a distribution of the domain and difficulty of the study material.

Conclusion

Our project shows that an RL based dynamic tutoring can

- **boost** student performance and
- **maintain** cognitive engagement.

Some aspects of the problem were out-of-scope for this project but would be useful to consider for **future work:**

- selecting RL methods with discrete action spaces and more resilient to variance
- using a larger dataset

References

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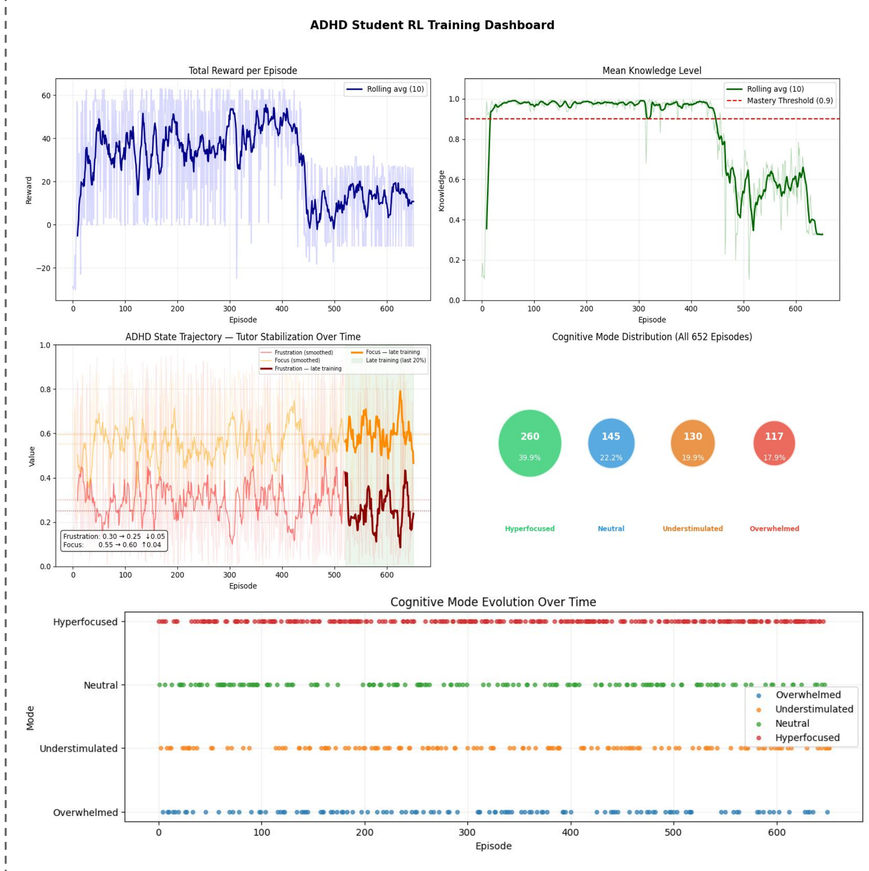
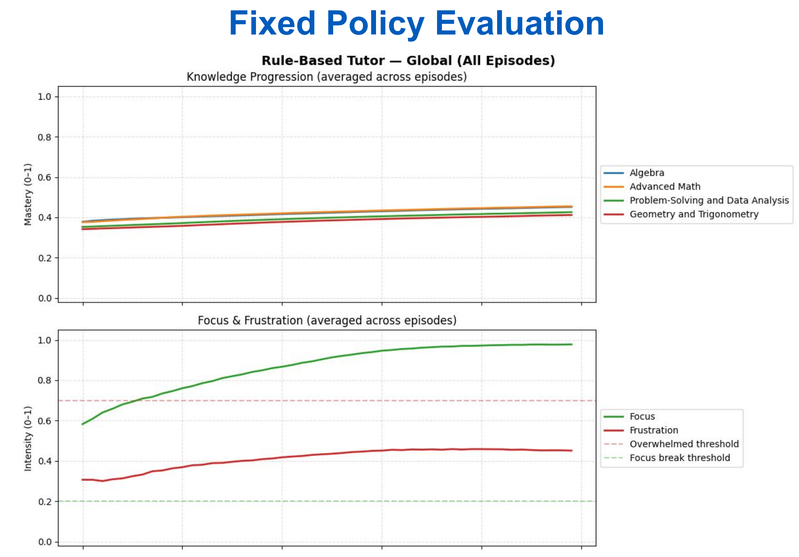
Results & Evaluation

Training with PPO algorithm

Training with DDQN algorithm



Fixed Policy Evaluation



Try it yourself!

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